

Sinusoda Juice is an advanced, hi-power, modular DC power adapter, designed for use with electronic music instruments.

It has a form of a single, multi-layer PCB unit, combining functions of PSU and busboard. Each Juice is equipped with a slim module (Sinusoda Straw) for connecting an external AC power adapter (optional) and a USB socket.

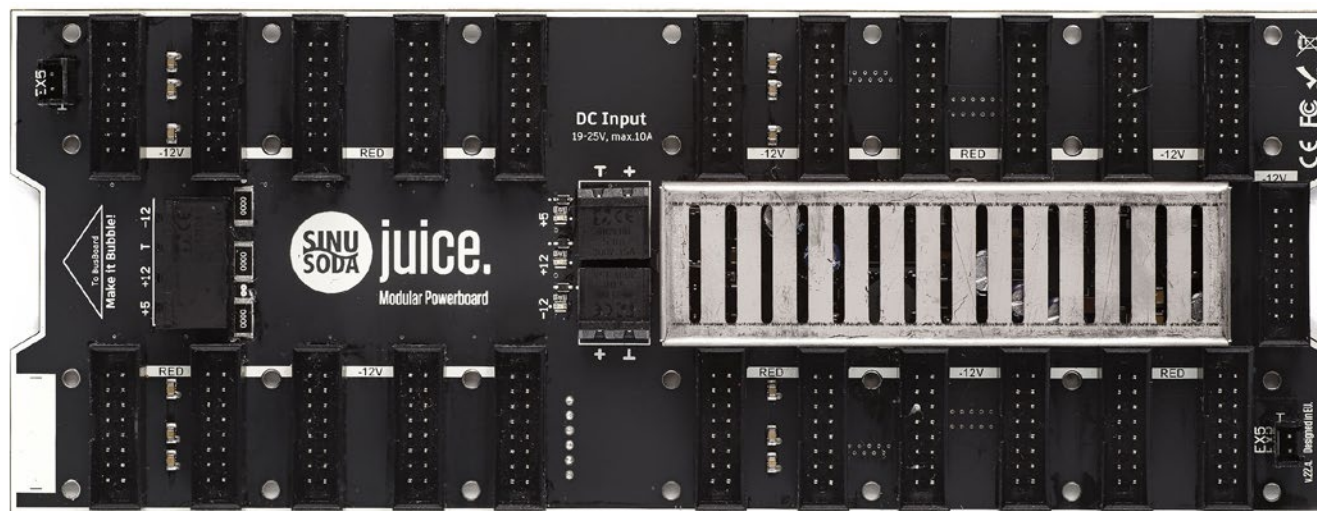


Figure 1. Juice - Modular Powerboard. Top view.

1. Features

- Input: DC 19 to 25V (nominal: 24V), up to 4 amps (10 amps pass-thru).
- Output: +12V @ 3A, -12V @ 1.5A, +5V @ 2A.
- 23 on-board headers for powering modular devices (2x8 pins, 0.1"/2.54mm pitch)
- 2 additional +5V headers (2 pins, 0.1"/2.54mm pitch) for external devices (max. load: 1400mA)
- Single header (EDG-type, 4 pins, 0.2"/5.08mm pitch) for connecting dedicated busboards: Sinusoda BUBBLES
- Wide expandability: possibility to connect up to two external busboards (Sinusoda Bubbles, each with 24 additional headers).
- Input connector: two EDG-type connectors (2 pins, 0.2"/5.08mm pitch) - one for input, another for passing-thru the input voltage to further devices (up to three Juice power supplies can be connected in series and powered from the single power socket).
- Input module (Sinusoda Straw) with power socket (max 25V/8A) and USB-A type power connector
- Hi-end output filtering to minimize noise and ripple
- Efficient EMC filter on input to reduce electromagnetic interference
- Steel shield over the electronics to reduce electromagnetic interference and enhance safety.
- High efficiency: up to 91% (depending on load characteristics)
- High tolerance to capacitive load
- LED indicators of proper voltage for each channel
- Safety marks (CE, FCC, ROHS), confirmed by third party tests and measurements
- Recommended AC power supply: Sinusoda Brick series (24V, min. 60W, max 200W).

2. Description

The Juice employs peak current control mode for immediate transient response and good line and load regulation. It is equipped with high-efficiency switching voltage regulators and fine-tuned, 2-stage filtering, to achieve a clean power supply, suitable even for the most demanding modular devices in the music industry.

The device represents a new concept in the design of modular PSU: the power supply is built together with the busboard to avoid interferences and noise. Thanks to its power reserve the compact PSU with 23 headers can be further expanded, by connecting additional dedicated busboards.

Up to two Sinusoda Bubbles busboards (each with 24 headers), can be connected to a single expansion socket (4-pin type), offering up to 71 headers for individual modular devices.

In addition, to ease the power distribution in large-scale modular instruments - it is possible to connect Juice devices in series, to reduce the number of external AC adapters. Due to the current limitations of DC connectors - the 10 amps power limit must be observed during connecting more than one Juice to a single AC adapter.

The Juice was designed to be used with Sinusoda modular housings, but it can be used also with other housings, available from different manufacturers. 28 mounting holes allow for easy installation in various environments. We recommend attaching the Juice to the housing with at least 14 screws, evenly distributed over the surface of the board. Screws should be always used with nylon washers, to avoid damaging the PCB and short-circuits.

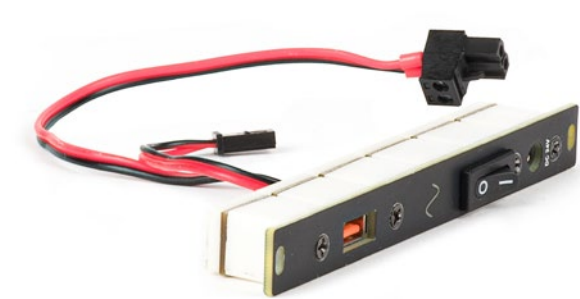


Figure 2. The Straw

Each Juice is equipped with a Sinusoda Straw - an elegant and compact module with a standard DC jack socket (2.5/5.5mm type), easy to use power switch, and a USB power socket, allowing to use of external devices (such as lamps) and charging the phone or tablet. The Straw has a form of a Eurorack module (3U/3HP) and can be easily mounted in the convenient part of the instrument’s housing.

Current limit of the Straw is 8 amps - please observe this limit while connecting additional Juices in series.

The Juice powerboard is safe and easy to use - both for the user and connected devices. Modern integrated circuits aboard are internally protected from overload, overheating, and short-circuit. In the

case of a short circuit - the device will simply switch off. After the problem is solved - the PSU will turn on again.

The metal shield covering the electronics reduces the interferences and reduces the risk of a short-circuit. Please do not remove it.

3. Technical Data, Electrical Characteristics and Operating Conditions

DESCRIPTION		MIN	TYP	MAX	UNIT
Supply voltage		19	24	25	V
Max. output current	for +12V channel		3		A
	for -12V channel		1.5		A
	for +5V channel		2		A
Input current limit (Straw module)			8		A
Voltage regulation			± 200		mV
Max capacitive load for each channel		10			mF
Output ripple (20MHz bandwidth, @ 50% of nominal load)	in +12V channel		14	30	mV p-p
	in -12V channel		29	60	mV p-p
	in +5V channel		15	30	mV p-p
Efficiency (@ 20-100% of nominal load)	in +12V channel	86	90	91	%
	in -12V channel	77	83	85	%
	in +5V channel	79	85	86	%
Operating temperature		5		40	°C
Dimensions (Juice)	length		226		mm
	width		86		mm
	height		19		mm
Dimensions (Straw)	length		128.5		mm
	width		15		mm
	depth		20		mm
Weight (Juice+Straw)		190	195	200	g

4. Juice Powerboard Efficiency

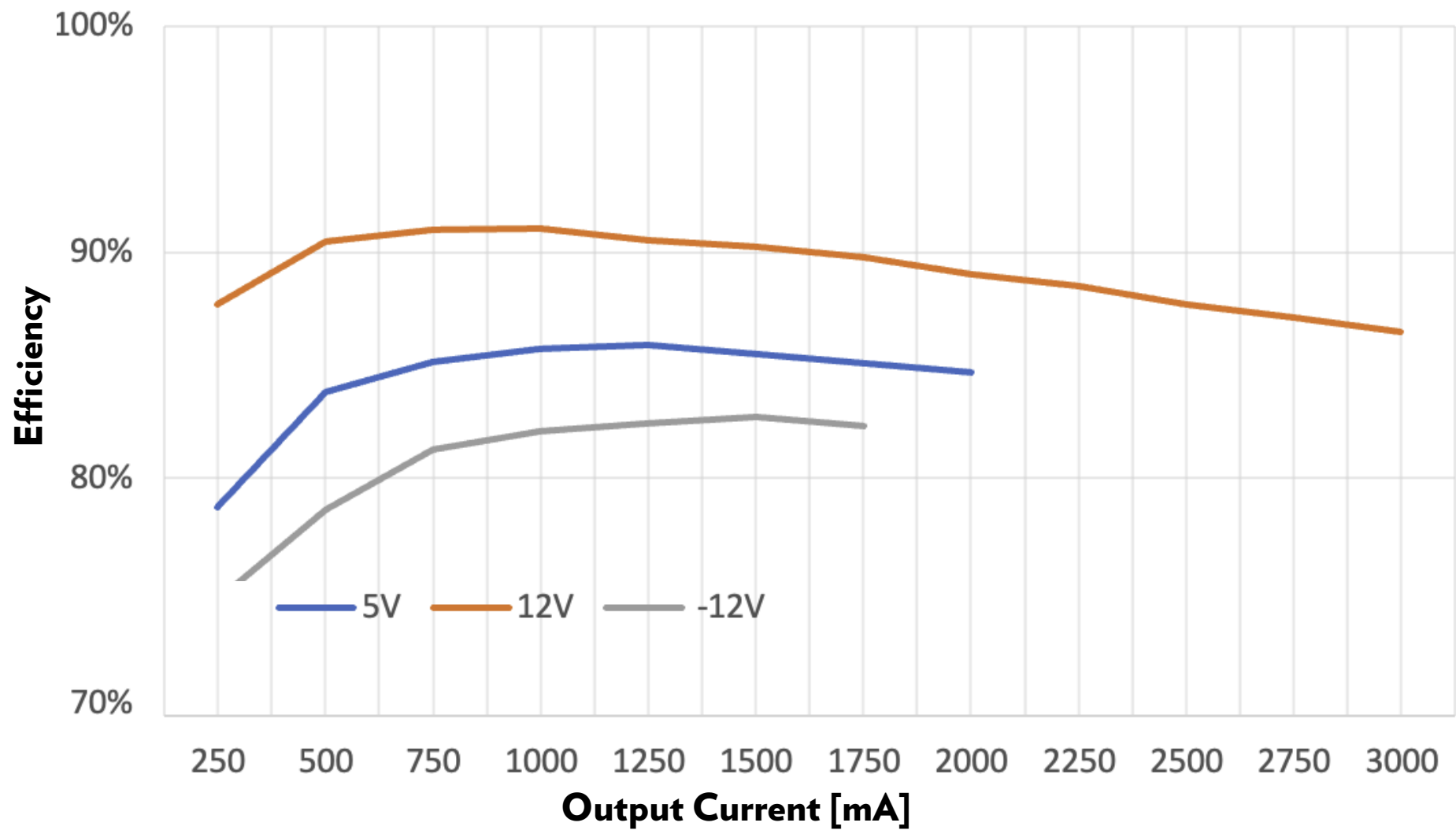


Figure 3. Juice Powerboard Efficiency

5. Output Noise & Ripple

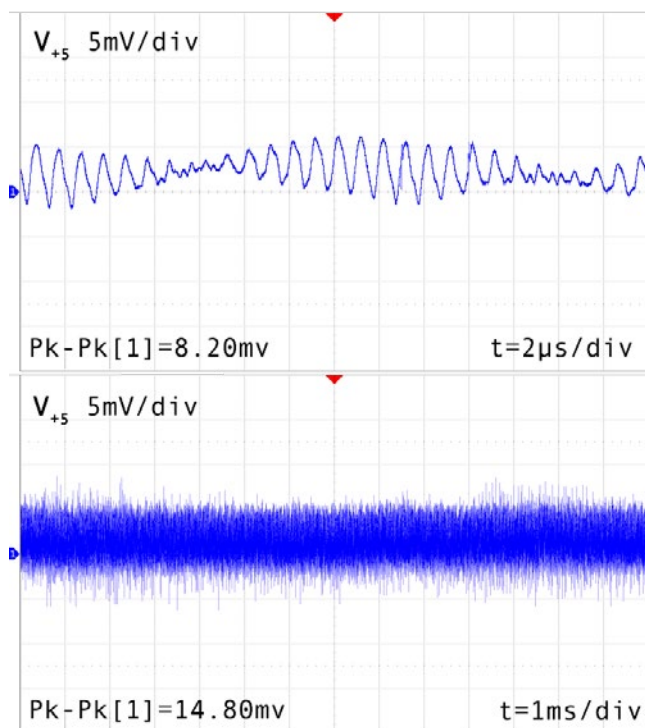


Figure 4. Output Noise & Ripple @ +5V Channel
Load = 1A

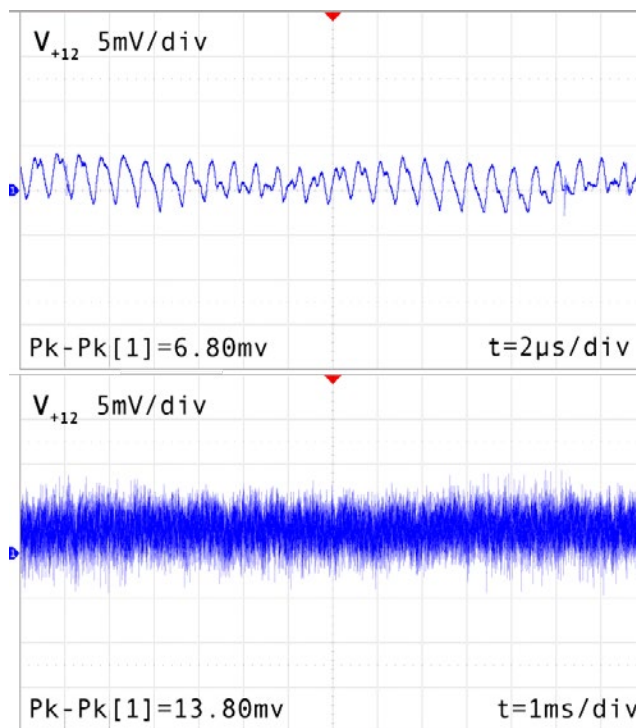


Figure 5. Output Noise & Ripple @ +12V Channel
Load = 1.5A

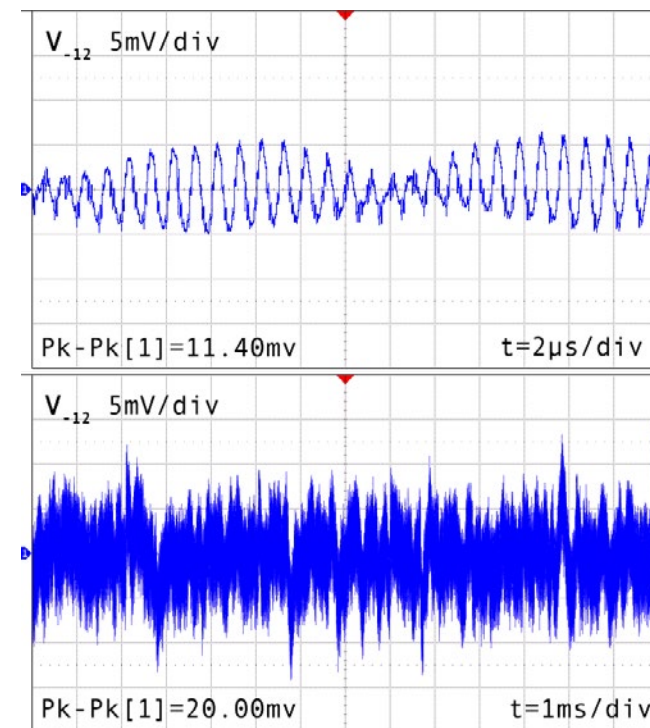


Figure 6. Output Noise & Ripple @ -12v Channel
Load = 1A



That's all folks!

Feel free to drop us a line on [**info@sinusoda.com**](mailto:info@sinusoda.com) if you have any questions, suggestions, or ideas or want to talk about gear :)